Exon-miRNA Classifier

Group 1

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Exon vs miRNA

Exons:

- Sequences of DNA (and later RNA) left after splicing of introns.
- Sequences that code for proteins.
- High variable in length. Can be few dozens long or 200 nucleotides long.
- Translated into proteins.

miRNAs:

- Small, completely non-coding RNA molecules.
- Regulates genes
- They attach themselves to mRNAs and then can either completely destroy the mRNA or block the ribosome reading it.
- Mature miRNAs are 22 nucleotides long.

Goal

Create a model that is trained on multi-species exon and miRNA data to predict whether a given DNA/RNA sequence is exon or miRNA.

Preprocessing

Choosing Dataset

We created our own dataset from different organisms because no such dataset is readily available. So we began choosing organisms based on the following criteria:

- Length of Genome
- Number of miRNAs
- Species variety

After careful evaluation many organism we concluded on the following 4 organisms which best suit our use case:

- Homo Sapiens (Humans)
- Danio rerio (Zebrafish)
- Mus musculus (mouse)
- Drosophila Melanogaster (fruit fly)

Dataset was downloaded from NCBI genome: https://www.ncbi.nlm.nih.gov/datasets/genome/

Class imbalance

In any organism exons outclass miRNAs by 1000x so we needed a strategy to reduce this number to overcome overfitting. There were 4 ideas we came up with:

1. Select first k exons
2. Select last k exons
3. Select k random exons (selected)
4. Select k exons across middle

We chose option 3.

Selecting k random exons because it gave us the diversity in the exon dataset that reduced the bias of the model.

We performed the selection for the 4 organisms and created the complete dataset.

Dataset Overview

Sequence	Label
TTGTGGAAGGGAAGGTCAACAAAG…	exon
TTGTGGAAGGGAAGGTCAACAAAG…	exon
CCTGAGGCAGAGAAGTGGAGAGAA…	miRNA
TGCTAGCCGATGCCCGCACTGTGC…	exon
…	…

Sequence Length

All most all the miRNA sequences are 22 nucleotides long but exons are 100-200 or more nucleotides long. If we feed this raw data to any model then the model will just learn to count the length and never understand its biological importance. So, we came up with 2 solutions:

- Truncation to 22 bp:
  - If sequence length > 22 then truncate to first 22 bp
  - If sequence length < 22 then add padding bps ('N' * (22-sequence length))
- Elongation to 100 bp:
  - Elongate the sequence from middle to increase its length to 100 bps.
  - Elongation is done to right and left side of the sequence.
- From experimentation we found out that Elongation yields better results.
- This might be due to that model has more data to generalize to the task.

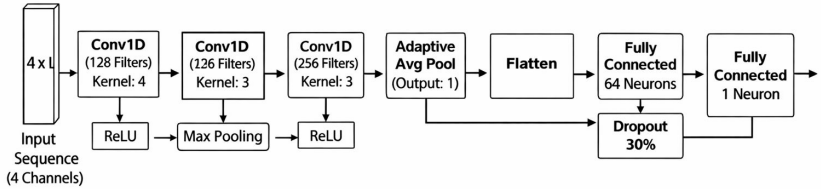
Model Selection

1D CNN

- 1D CNNs are standard baseline for any task related to sequence data.
- 1D convolution acts like positional weight matrices or motif finders, scanning the sequences for short, specific nucleotide patterns.
- Advantages:
  - Speed & efficiency - Highly parallelizable
  - Motif detection - Identifying local features
  - Low risk of overfitting - smaller parameter count
- Disadvantages:
  - Limited context - small context window

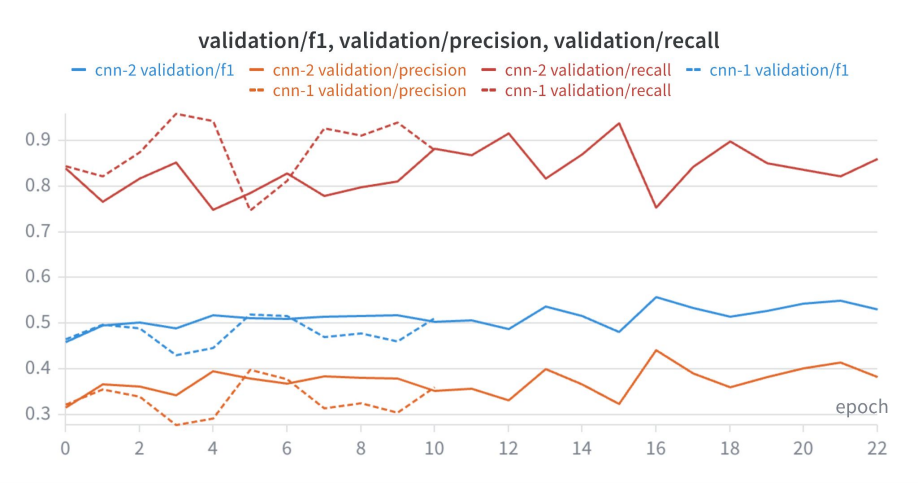
1D CNN Architecture

```
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================
RNACnn                                   [64, 1]                   --
├─Conv1d: 1-1                            [64, 128, 97]             2,176
├─MaxPool1d: 1-2                         [64, 128, 48]             --
├─Conv1d: 1-3                            [64, 256, 46]             98,560
├─AdaptiveAvgPool1d: 1-4                 [64, 256, 1]              --
├─Flatten: 1-5                           [64, 256]                 --
├─Linear: 1-6                            [64, 64]                  16,448
├─Dropout: 1-7                           [64, 64]                  --
├─Linear: 1-8                            [64, 1]                   65
```



1D CNN Results

- Recall is high that means our model captures most of the miRNAs.
- Precision is low which indicates that in those captured miRNAs very few were actually miRNAs.
- F1 score is around 0.5 indicating that model is always 50% right about its decision.

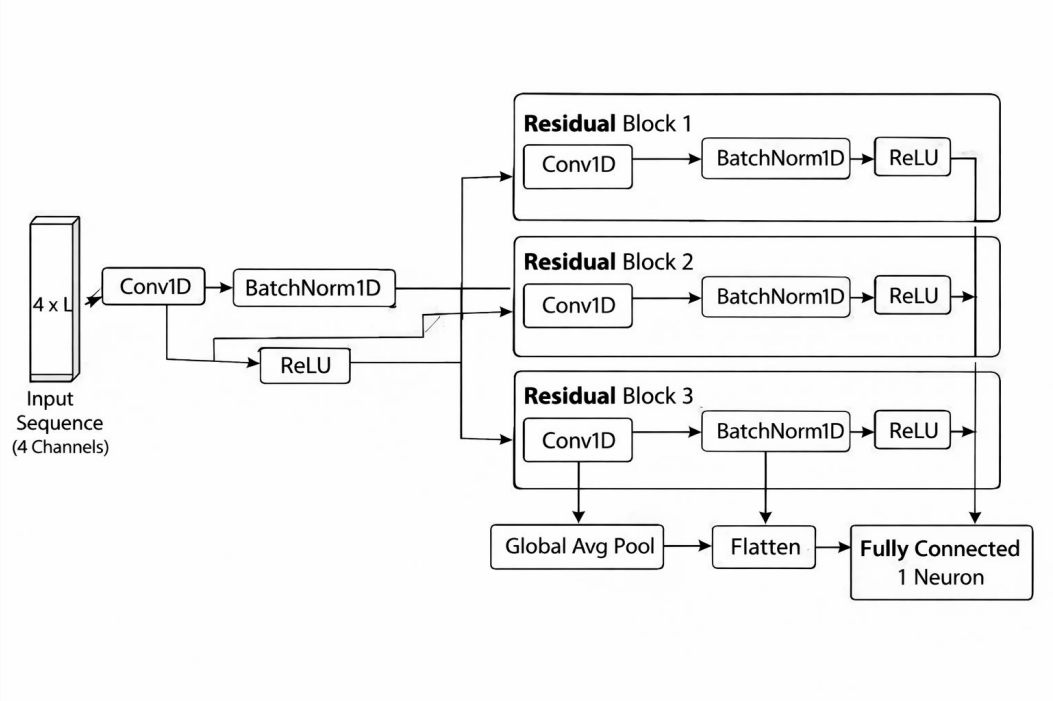


1D Resnet

- Makes a 1D CNN deeper without suffering vanishing gradient.
- Allows the model to learn hierarchy of features, from simple motifs to complex secondary structure indicators.
- Advantages:
  - Deeper receptive field - wider spatial window
  - Stable training - train a deeper and accurate model
  - Hierarchical Representation - higher-level sequence representations
- Disadvantages:
  - Heavy Computation - slower to train
  - Localized - lacks memory of distant past

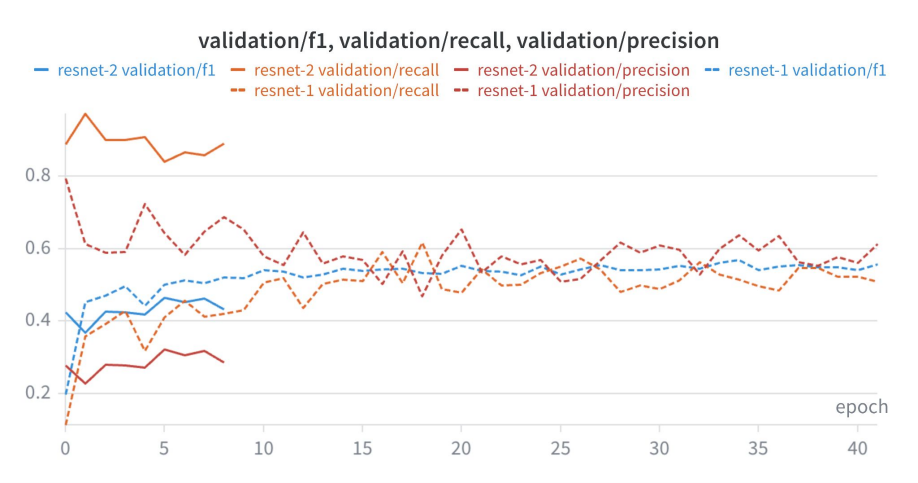
1D Resnet Architecture

```
==========================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================
RNAResNet1D                              [64, 1]                   --
├─Conv1d: 1-1                            [64, 64, 100]             1,280
├─BatchNorm1d: 1-2                       [64, 64, 100]             128
├─ReLU: 1-3                              [64, 64, 100]             --
├─ResidualBlock1D: 1-4                   [64, 64, 100]             --
│    └─Sequential: 2-1                   [64, 64, 100]             --
│    └─Conv1d: 2-2                       [64, 64, 100]             12,288
│    └─BatchNorm1d: 2-3                  [64, 64, 100]             128
│    └─ReLU: 2-4                         [64, 64, 100]             --
│    └─Dropout: 2-5                      [64, 64, 100]             --
│    └─Conv1d: 2-6                       [64, 64, 100]             12,288
│    └─BatchNorm1d: 2-7                  [64, 64, 100]             128
│    └─ReLU: 2-8                         [64, 64, 100]             --
├─ResidualBlock1D: 1-5                   [64, 128, 50]             --
│    └─Sequential: 2-9                   [64, 128, 50]             --
│    │    └─Conv1d: 3-1                  [64, 128, 50]             8,192
│    │    └─BatchNorm1d: 3-2             [64, 128, 50]             256
│    └─Conv1d: 2-10                      [64, 128, 50]             24,576
│    └─BatchNorm1d: 2-11                 [64, 128, 50]             256
│    └─ReLU: 2-12                        [64, 128, 50]             --
│    └─Dropout: 2-13                     [64, 128, 50]             --
│    └─Conv1d: 2-14                      [64, 128, 50]             49,152
│    └─BatchNorm1d: 2-15                 [64, 128, 50]             256
│    └─ReLU: 2-16                        [64, 128, 50]             --
├─ResidualBlock1D: 1-6                   [64, 256, 25]             --
│    └─Sequential: 2-17                  [64, 256, 25]             --
│    │    └─Conv1d: 3-3                  [64, 256, 25]             32,768
│    │    └─BatchNorm1d: 3-4             [64, 256, 25]             512
│    └─Conv1d: 2-18                      [64, 256, 25]             98,304
│    └─BatchNorm1d: 2-19                 [64, 256, 25]             512
│    └─ReLU: 2-20                        [64, 256, 25]             --
│    └─Dropout: 2-21                     [64, 256, 25]             --
│    └─Conv1d: 2-22                      [64, 256, 25]             196,608
│    └─BatchNorm1d: 2-23                 [64, 256, 25]             512
│    └─ReLU: 2-24                        [64, 256, 25]             --
├─AdaptiveAvgPool1d: 1-7                 [64, 256, 1]              --
├─Linear: 1-8                            [64, 1]                   257
==========================================================================
```



1D Resnet Results

- Recall, precision and f1 score eventually converges to around 0.5.
- This indicates that the model is 50% sure that it has captured all the miRNAs and 50% sure that in those there are actual miRNAs.
- It is basically flipping a coin and deciding the result.



BiLSTM

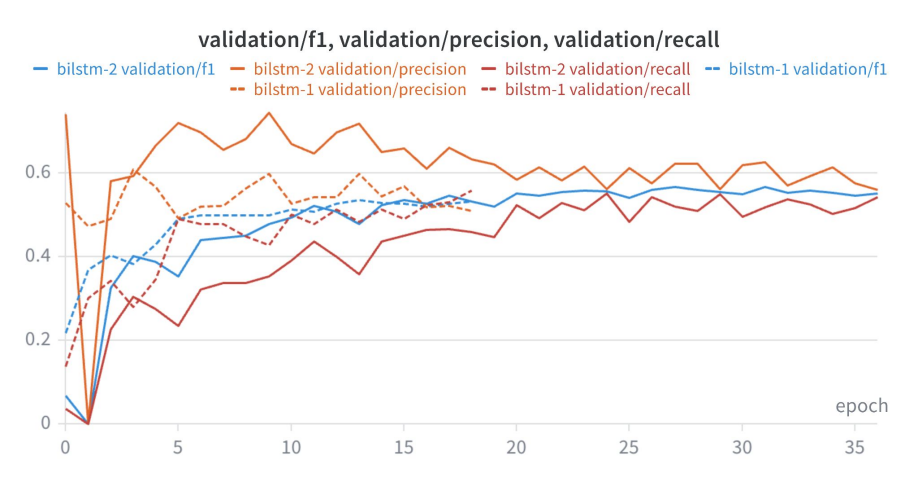
- Bidirectional Long Short-Term Memory networks are specially designed for sequential data where the order of the “tokens” strongly dictates the meaning.
- In RNA, the upstream and downstream context are very useful.
- Advantages:
  - Long-Ranged dependencies - remember information from distant past
  - Bidirectional context - can read sequence from both 3’ - 5’ and 5’ - 3’
  - Variable lengths - No fixed sequence length is needed
- Disadvantages:
  - Slower than previous models - process tokens sequentially
  - Memory intensive - stores many hidden states
  - Overfitting

BiLSTM Architecture

```
==========================================================================
Layer (type:depth-idx)                   Output Shape              Param #
==========================================================================
RNABiLSTM                                [64, 1]                   --
├─LSTM: 1-1                              [64, 100, 256]            532,480
├─Linear: 1-2                            [64, 32]                  8,224
├─Dropout: 1-3                           [64, 32]                  --
├─Linear: 1-4                            [64, 1]                   33
==========================================================================
```

BiLSTM Results

- Similar results to 1D Resnet but recall, f1 and precision seem to converge around 0.6 which is a better result than the previous model.
- It is better than just flipping a coin and choosing.
- Trains faster and gives better results than 1D Resnet.



XGBoost (Ensemble Decision Tree)

- Each tree trained to predict and correct residual errors of the previous trees.
- Uses L1/L2 regularization to prevent overfitting
- Highly optimized for performance and speed.

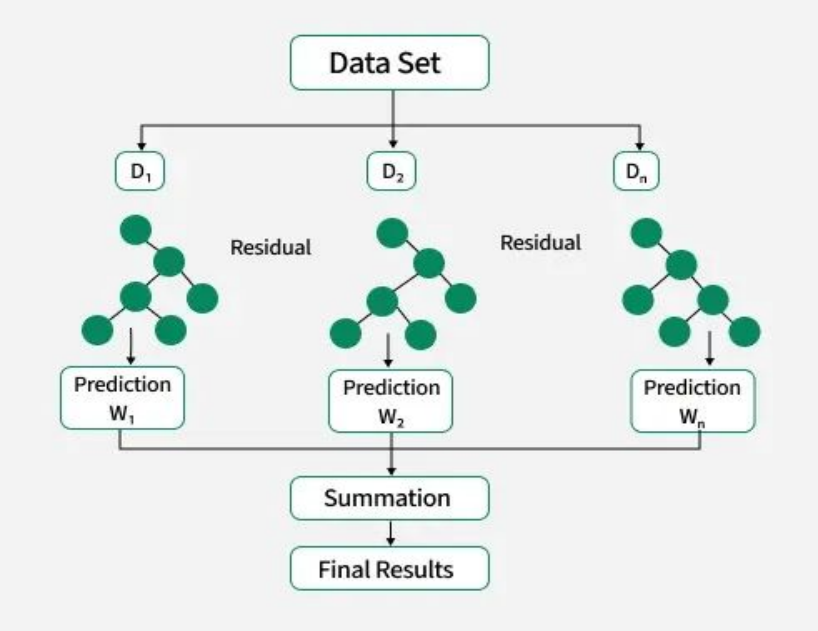


Image source: https://media.geeksforgeeks.org/wp-content/uploads/20250905153324093033/data_set.webp

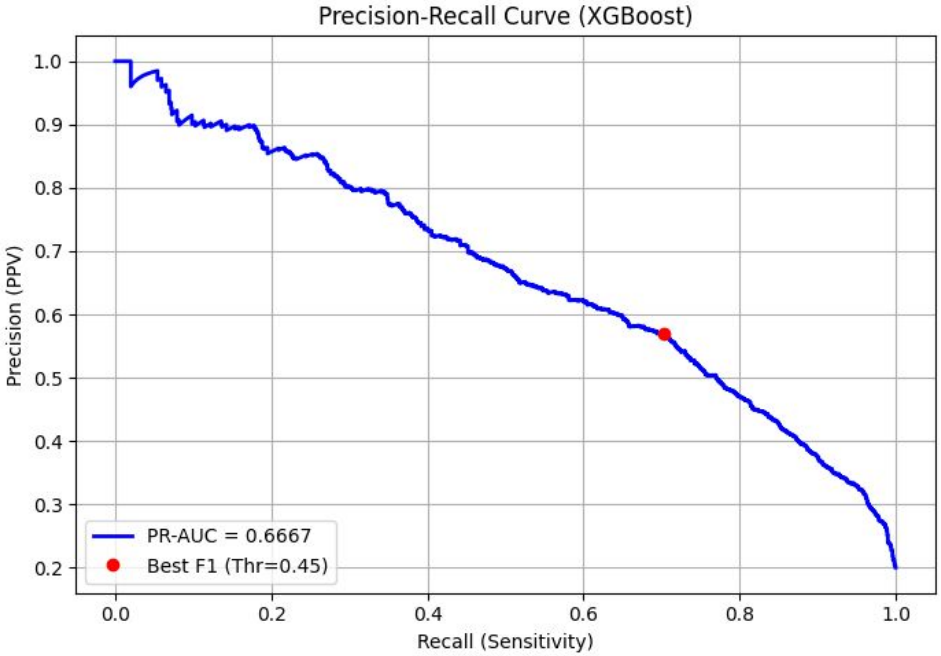
XGBoost

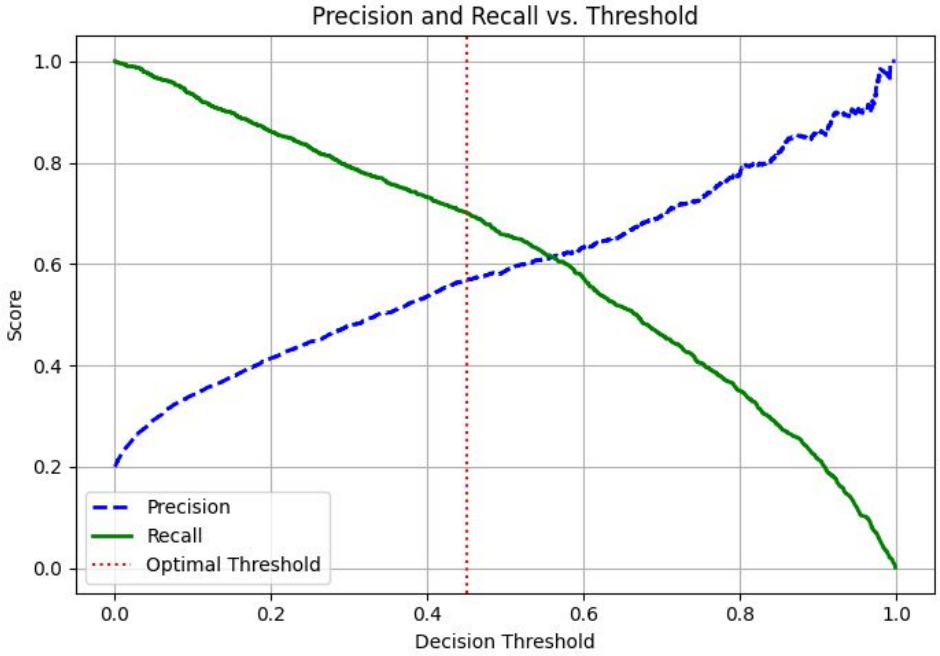
- Feature Engineering:
  - Frequencies of K-mers (2-mers to 5-mers)
  - GC content percentage
- Built-in Imbalance Handling:
  - It allows direct tuning of the scale_pos_weight parameter, powerfully penalizing false -ves for miRNA class.
- Interpretability:
  - Provides explicit feature_importances_ allowing to verify which specific k-mers or features are biologically driving the classification.
- Hyperparameter Tuning:
  - Pairs directly with systematic search algorithms and allows for precise decision-threshold tuning.

Best parameters:

```
{
'subsample': 0.9,
'scale_pos_weight': 5.0,
'n_estimators': 500,
'max_depth': 4,
'learning_rate': 0.2,
'gamma': 0,
'colsample_bytree': 1.0
}
```

XGBoost Result





XGBoost Result

- Recall of 0.7 means it captures 70% of the miRNAs.
- Precision of 0.57 means among the captured miRNAs 57% are correctly identified.
- A f1 score of 0.63 means 63% of the times model classifies correctly.
- Best result among all models.

```
=> Optimal Decision Threshold: 0.4498 (Max F1: 0.6288)
```

	precision	recall	f1-score	support
Exon (0)	0.92	0.87	0.89	4798
miRNA (1)	0.57	0.70	0.63	1199
accuracy			0.83	5997
macro avg	0.75	0.78	0.76	5997
weighted avg	0.85	0.83	0.84	5997

Conclusion

- Feature Engineering Prevailed:
  - Methods like k-mers and GC content proved to more effective than forcing DL models to infer representation.
- Simplicity Beats Complexity:
  - Tradition XGBoost crushed deep PyTorch architectures (CNN, Resnet1D, BiLSTM), showing that strong domain features beat deep networks.
- Algorithmic Imbalance Handling:
  - Tuning scale_pos_weight and shifting the decision threshold based on precision-recall curves successfully neutralized the dataset’s heavy 4:1 class imbalance.

Thank You